



COSMO INSTRUMENTS (I) PVT. LTD.
SPECIALIST IN AIR LEAK AND FLOW TEST



AIR LEAK TESTER

MODEL LS-1866

ERROR AND TROUBLESHOOTING MANUAL

1 Error Messages

Errors are detected in either the power-on check procedure, or during execution of a leak test. If an error is detected, error code will be displayed on the leak rate indicator along with blinking No-Go LED and beep sound. An ERROR signal will be transmitted through the control I/O port. Pressing **STOP** key clears the error display and signal.

1.1 Errors and Treatment

Sr. No.	Error Code	Error Message	Treatment
1	E-1	Flash memory error	Replace the CPU board.
2	E-2	DPS Oscillation stop	Sensor electric source, cable disconnection, electric circuit damage. Repair by manufacturer required.
3	E-3	Excessive DPS zero-point shift	Check and adjust the stop zero point. (See Section 1.2.)
		Water, oils, or contamination in DPS	After checking and adjusting the stop zero point, repeat the charge hold to check the zero point several times. If the stop zero point tends to shift or lacks repeatability, DPS may be contaminated. Have a Cosmo representative replace DPS. Detected in DL1 stage.
4	E-4	Improper test pressure	Test pressure may be out of range of the pressure switch. Check the test pressure switch settings on the gauge. Adjust the test pressure or reset the test pressure limits. (See Section 1.3.) Detected at the end of CHG stage.
5	E-5	Air-operated valve inactive	Verify that the pilot pressure is regulated at least 400 kPa. (See Section 1.4.) Detected in CHG or DL3 stage.
6	E-6	A/D Converter error	Exchange of CPU board.
7	E-7	DPS output saturated	Check the DPS offset. No-Go judgement will be made. ERROR signal won't be transmitted. Detected at the end of DET stage.
8	E-8	I/O Output error	Cable disconnection, electric circuit damage. Repair by manufacturer required.
9	E-9	Pressurization valve error	Pressurization valve AV1 not closed. Air operated valve AV1, pilot pressure solenoid valve error. Repair by manufacturer required.

The LO NO GO LED will light when a static memory (SRAM) error is detected. In such case the CPU board must be replaced.

1.2 E-3: Excessive Differential Pressure Sensor Zero-point Shift

When leak test is repeated without unclamping, the excessive DPS zero-point shift may occur. This tends to occur when the volume of the tested part is so large or the tubing is so narrow that the air inside the tested part is not completely exhausted, resulting remaining pressure is detected during DL1 stage in the following test. In such case, delay the input of the next START signal or extend the DL1 timer to let the tester exhaust completely. The monitoring function may also be disabled with a memory switch.

1.3 E-4 Improper test pressure

When arranging a pressure switch to WORK-side measurement system and not using the original one, E-4 error message might appear. In that case, the wrong setting of the test pressure or malfunction of the air-operated valve could be its cause.

NOTE: When arranging a pressure switch to work side measurement system, the effect of the volumetric change of pressure switch could cause the reduced sensitivity and leak.

1.4 E-5: Air-operated Valve Inactive

When the error E-5 is indicated, check the following:

1) **Test pressure**

A positive pressure of 8 kPa or more or a negative pressure of -13 kPa or more is required.

2) **Pilot pressure**

A pilot pressure should be regulated 400 to 700 kPa.

3) **DL3 timer**

A low test pressure model will require longer DL3 timer than usual. Extend the DL3 timer by 1 to 2 seconds. In case of Micro-low pressure model may require optional orifice.

4) **Air-operated valves and solenoid valves**

If contaminants are found in the air-operated valves, cleaning or replacement is required.

5) **DPS sensitivity**

Check the DPS sensitivity using a Leak Master, and DPS span.

1.5 E-9 Pressurization valve error

In case WORK & MASTER volume are largely different or exhaust time takes longer, the offset of the differential pressure sensor could exceed its limit and display E-9 error message. In those cases, please arrange the time for more than 5 sec by memory switch.

2 Leak Test Troubleshooting

2.1 When No-Go judgments occur consecutively

No-Go judgment		Probable cause	Treatment
Continuous No-Go	Tested part	Internal mold cavity of the tested part. Leaks from sealing caused by machining defects	Perform charge hold applying soap solution on the sealing, and or water-immersion test.
		Volume change caused by part and/or internal seal deformation	Change the timer settings or test pressure.
		Differential pressure due to temperature changes	Let the part be ambient temperature.
	The sealing fixture, tubing or other parts	Worn-out or contaminated sealing	Clean or replace sealing as required.
		Leak from the tubing or valves	Perform charge hold applying soap solution on the tubing and valves.
		Differential pressure due to seal volume changes caused by clamping pressure	Implement exhaust interference prevention to guard against effects from other testers if any. Review the sealing fixture and rubber designs.
		Temperature changes of the fixture after water-immersion	Blow the water away and control the water temperature.
	Tester	Internal leak or contamination of DPS	Perform No Leak test with the stop valves closed.
		Improper test timer	Execute Master-Preset sampling with non-leaking part, and check if the value is within the limit. If it is, extend timers.
		Improper equivalent internal volume	Perform leak calibration with a Leak Master calibrated to specific leak rate.
	Master-Preset Compensation	Improper tested part (leak, deformation or temperature effects). The part used for Master-Preset value sampling is not actual part.	Use actual non-leaking tested part for sampling Master-Preset value.
When the BAL stage completes, the NG occurs.		Improper upper or lower limit test pressure	Verify the upper and lower limit settings.
		Large leak	After checking the sealing rubber for wear or stuck foreign matter, replace with a conforming tested part.
		Improper test pressure	Check the pressure source. Verify the regulator setting.

The Master-Presetting results in an NG judgement.		Leak from the tested part, etc.	Retry Master-Preset with another part.
		Temperature change in the tested part	Do not carry out Master-Preset with a special part, such as one that has just been subjected to water-immersion checking.
The judgement of master NG occurs frequently.		Leak from the master	Replace the master.
		Poor master temperature stability	Replace with a Master Chamber having good temperature stability.

3 Locating Leaks in Places Other Than the Leak Tester

- a) Press the CHG HOLD key to pressurize the tested part and locate leaks applying soap solution.

NOTE: Do not apply soap solution to a vacuum pressure type tester.

Disconnect the tubing and fixture from the WORK-side port and apply regulated micro-low pressure and soap solution to locate the leak.

- b) Execute a leak test by switching the master and tested part.

If a tested part was rejected but the leak can't be located even by applying soap solution, switch the tubing and fixtures of the MASTER-side and WORK-side. This helps find out whether the leak is from the MASTER-side or WORK-side tubing or fixture.

If the polarity of the leak rate changes when tubing and fixtures of the MASTER and the WORK are switched, a leak is more likely to locate in the tested part or fixture seal. Do this check with the Master-Preset feature disabled.

Be sure to replace the tubing in original position when the check is completed.

NOTE: If soap solution is used, be sure to wipe it off with a cloth.